CHUAN ZOU

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RESEARCH INTERESTS

I am interested in **3D perception**, **robot learning**, and **AI agents** for robots that operate in **construction and the built environment**.

EDUCATION

University of Pennsylvania *2024 – 2026*
Master of Science, Stuart Weitzman School of Design GPA 3.67 / 4.0

- Relevant coursework: Programming Languages and Techniques (CIT 5900) and Principles of Deep Learning (ESE 5460) in Penn Engineering; Deep Learning with Python; Structures I–II.

Tongji University *2018 – 2024*
Bachelor of Engineering, College of Architecture and Urban Planning GPA 3.93 / 4.0 (WES)

- First year in Mechanical Engineering (honor student, departmental scholarship); transferred to the Architecture Honors Program and restarted from year one.
- Minor in French Language, including a one-year intensive program (2022–2023).
- Relevant coursework: Advanced Mathematics, Theoretical Mechanics, General Physics, Python Programming.

SELECTED PUBLICATIONS

EvolvAttack: Backdooring Self-Evolving VLM Agents
C. Zou, Y. Xiang, Z. Li, X. Luo, S. Chen†
Under review; arXiv:2610.15244, 2026

- Identified the self-evolving backdoor, whose payload stays dormant at deployment and activates only once the agent's own memory has evolved; EvolvAttack stays nearly dormant before evolution (0.5% ASR) yet exceeds 95% ASR within 100 interactions across three VLM backbones and four agent pipelines.

StateHarness: Maintaining Evolving Project State for Long-Horizon Coding Agents
C. Zou, Y. Zhou, X. Luo, X. Zhang†
Under review; arXiv:2610.17616, 2026

- Proposed a training-free runtime that keeps an explicit project state for long-horizon coding agents and retrieves the callers, tests, and documentation each edit affects; improved RefactorBench full-chain completion from 29.4% to 35.3% and LHTB mean reward from 0.427 to 0.576.

† Corresponding author

PUBLICATIONS INVOLVED

- X. Luo, Y. Huang, K. Guo, P. He, **C. Zou**, C. Gao, L. Li, Y. Ma, Z. Xu, Z. Chen, Y. Han, X. Zhang. "SecProbe: Adaptive Evaluation of Coding Agents on Cybersecurity Vulnerabilities," *Under review; arXiv:2609.33763*, 2026
- X. Luo, Y. Huang, K. Guo, P. He, **C. Zou**, T. Hua, X. Zhang. "Learning from Synthetic Data without Model Collapse in Iterative Instruction Tuning," *Under review; arXiv:2607.17043*, 2026

RESEARCH EXPERIENCE

George Mason University Aug 2026 – Oct 2026
Research Intern; first-author project: EvolvAttack (see Publications) Fairfax, VA, USA

University of Notre Dame Jun 2026 – Aug 2026
Research Intern; first-author project: StateHarness (see Publications) Notre Dame, IN, USA

Texas A&M University Aug 2025 – Dec 2025
Computer Vision and AI Intern, Civil Engineering × Robotics Group USA
Project: National Cooperative Highway Research Program (short exploratory project, unpublished)

- Built a pipeline that reconstructs a work zone in 3D from a few dash-cam frames (VGGT), detects traffic cones with a fine-tuned YOLO model, lifts them into 3D, and recovers metric scale with PnP.
- Predicted dense depth from high-speed dash-cam video with DPT, and explored LoRA on frozen VGGT attention layers to stabilise depth under weather-induced domain shift.

Center for Satellite Application Technology of Natural Resources Jun 2025 – Aug 2025
ML Intern Changsha, China

- Cleaned noisy and inconsistent samples for CNN-based land classification and engineered spectral–temporal features (PCA, tree-based models), improving training stability.

PROJECTS

Generative Models for Cloud Removal in Satellite Imagery Spring 2026
Course project (MUSA 650), project lead, team of three University of Pennsylvania

- Built a controlled benchmark on 2,000 Sentinel-2 tiles of Beijing with synthetic clouds at 5–70% coverage to find where generative fill-in stops being reliable for land-use and urban monitoring.
- Compared a conditional GAN and mask-guided diffusion against multi-temporal fusion; fusion suffices below ~30% coverage, above which the GAN is most stable.

CONSTRUCTION AND FABRICATION

Knitted Soft Concrete Pavilion, DigitalFutures Core Studio, Tongji University Jul 2023
Team of 16; design, construction, and documentation

- Built a full-scale concrete shell cast on knitted textile formwork over CNC-cut plywood frames.

ZUO Architecture (Original Design Studio, TJAD) 2023, 5 months
Architectural Intern Shanghai, China

- Built 3D models and prepared construction drawings for four built public projects in Taizhou, Shenzhen, and Shanghai, including a 108-metre slow-traffic bridge and a metro station.

OTHER EXPERIENCE

Sand AI, *Product Manager Intern, Generative Video Agent (VidMuse)*, Beijing Jan 2026 – Present

SKILLS

Programming and ML	Python, Java, SQL; PyTorch, Transformers, verl, LoRA/PEFT, Git
Design and Survey	Rhino, Grasshopper, SketchUp, BIM, AutoCAD, Reality Capture